

BEREZOVO, Russian Federation

WMO: 236310

Lat: **63.933N**Lon: **65.050E**Elev: **32**StdP: **100.94**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.5	-37.8	-43.0	0.1	-40.3	-40.2	0.1	-37.7	9.6	-12.1	8.8	-13.6	1.2	60	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.0	19.5	24.9	18.4	22.9	17.3	20.7	25.1	19.5	23.5	18.1	21.9	3.5	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	23.1	17.8	12.8	22.1	16.5	11.8	20.7	59.9	25.3	55.8	23.4	51.5	21.9	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.8	8.9	7.8	-42.5	29.8	3.8	1.6	-45.3	31.0	-47.5	31.9	-49.6	32.8	-52.4	33.9	
(5)				-42.4	22.3	3.5	1.2	-44.9	23.2	-46.9	23.9	-48.8	24.6	-51.3	25.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.8	-21.5	-19.9	-10.3	-3.2	4.5	12.6	16.3	12.3	6.6	-1.2	-12.9	-18.4
	DBStd	14.67	9.11	8.88	7.54	6.64	5.71	4.96	4.25	3.83	4.09	5.58	8.77	9.71
	HDD10.0	5115	976	837	631	398	191	29	3	18	118	348	688	879
	HDD18.3	7767	1234	1070	889	647	429	182	91	191	353	605	938	1138
	CDD10.0	432	0	0	0	1	20	107	199	89	16	1	0	0
	CDD18.3	43	0	0	0	0	1	9	29	4	0	0	0	0
	CDH23.3	358	0	0	0	0	11	81	233	32	1	0	0	0
	CDH26.7	64	0	0	0	0	1	14	44	6	0	0	0	0
Wind	WSAvg	3.4	2.8	3.0	3.4	3.8	4.1	3.9	3.4	3.2	3.3	3.4	3.1	2.9
Precipitation	PrecAvg	523	27	23	22	33	43	61	69	72	59	51	36	28
	PrecMax	727	53	63	49	67	115	189	163	164	112	112	100	64
	PrecMin	393	10	4	3	8	16	18	8	18	8	22	4	7
	PrecStd	83	10	12	13	15	24	35	40	35	28	20	17	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	-0.1	6.1	14.5	25.1	28.5	29.7	26.7	21.0	13.0	3.2	0.0
		MCWB	-1.5	-1.3	1.8	8.1	15.1	18.9	21.0	19.7	15.8	9.9	2.1	-0.8
	2%	DB	-3.7	-2.5	3.6	10.1	21.3	25.6	27.9	23.5	17.3	10.0	1.7	-1.5
		MCWB	-4.2	-3.5	0.6	5.3	13.1	17.7	20.6	18.2	13.3	7.7	0.9	-2.1
	5%	DB	-6.2	-4.7	1.7	7.6	17.7	23.2	25.9	21.1	15.0	7.8	0.0	-3.4
		MCWB	-6.9	-5.5	-0.6	3.8	11.2	16.5	19.6	16.7	12.0	6.1	-0.6	-4.0
	10%	DB	-8.6	-6.7	-0.1	5.5	14.1	21.0	24.1	19.2	12.9	5.7	-1.5	-5.7
		MCWB	-9.1	-7.5	-1.8	2.4	9.2	15.5	18.7	15.6	10.6	4.5	-2.1	-6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-0.9	2.4	8.5	15.7	20.3	23.0	21.0	16.5	10.7	2.5	-0.8
		MCDB	-0.9	-0.2	5.5	13.7	23.0	25.2	27.9	25.9	20.1	12.5	3.0	-0.3
	2%	WB	-4.2	-3.4	0.9	5.8	13.7	18.9	21.5	19.0	14.1	8.2	0.9	-2.1
		MCDB	-3.7	-2.5	3.0	9.3	20.3	24.0	26.0	22.4	16.5	9.7	1.7	-1.6
	5%	WB	-6.8	-5.4	-0.3	4.1	11.7	17.4	20.4	17.4	12.3	6.2	-0.5	-4.0
		MCDB	-6.3	-4.7	1.5	7.1	16.6	22.1	24.6	20.2	14.3	7.7	0.0	-3.4
	10%	WB	-9.1	-7.5	-1.9	2.7	9.7	15.9	19.3	16.0	10.9	4.6	-2.2	-6.1
		MCDB	-8.6	-6.8	-0.1	5.1	14.3	20.3	23.3	18.6	12.7	5.6	-1.5	-5.6
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.3	9.8	9.5	9.1	8.8	9.2	8.7	7.3	5.7	6.6	7.3
		MCDBR	7.4	7.9	9.0	11.0	14.5	12.6	12.1	12.0	10.2	7.5	5.4	7.4
		MCWBR	7.2	7.2	7.0	7.3	8.1	6.4	5.9	6.9	6.6	5.6	5.2	7.2
	5% WB	MCDBR	7.3	7.8	8.9	9.9	13.4	10.7	10.5	10.5	9.2	7.0	5.5	7.1
		MCWBR	7.2	7.2	7.2	6.9	7.9	6.1	5.6	6.5	6.5	5.5	5.4	6.9
Clear-Sky Solar Irradiance	taub	0.221	0.280	0.325	0.354	0.362	0.380	0.422	0.390	0.338	0.301	0.221	0.223	
	taud	1.922	2.039	2.034	2.104	2.349	2.402	2.268	2.328	2.445	2.302	1.963	1.902	
	Ebn at Noon	333	609	736	809	848	838	782	774	751	604	323	114	
	Edh at Noon	30	68	104	123	106	103	115	99	72	53	27	13	
All-Sky Solar Radiation	RadAvg	0.19	0.96	2.57	4.34	5.17	5.33	5.29	3.58	2.02	0.85	0.26	0.07	
	RadStd	0.01	0.07	0.13	0.46	0.39	0.37	0.52	0.34	0.21	0.11	0.04	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page